(a) 16-bit formats and their limits

fp32 exp 8 | mantissa 23 — max ≈ 3.4×10^{38}

bf16 exp 8 | mant. 7 — max ≈ 3.4×10^{38}

fp16 exp 5 | mantissa 10 — max = 65,504

(b) SDXL VAE decoder, same latent

